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Final project proposal. **Novel visualization techniques for git data**.

0.1 Problem description

In this project we will create several novel visual analytic systems for git data. The tools we develop will help developers quickly gain an understanding of git repositories. We will focus on a few key questions.

1. Who are the main contributors to a repository
2. At what times was development most active for any repository
3. Can we identify times when AI assisted development because popular on a given repository
4. What is the best visualization for giving developers a overview of the project structure

0.2 Problem description and motivation

Developers often work with large amounts of code and are forced to quickly learn the custom's and conventions of many repositories, this process can be very time-consuming. By giving developers tools for quickly visualizing the history of a repository, we can immediately give them an understanding of conventions, key contributors, and etiquette.

Many visualizations have been developed over the years to work with git data, in this project we want to develop a handful of new visualizations by first studying visualizations that other have developed and refining them to improve there visual clarity, augmenting them with additional data or even combining two visualization techniques into one.

Currently we have planned 3 visualizations but may add more or modify some of them over the course of the project

1. A 'stabilized' Sediment Graph based on the work Erik Bernhardsson. Erik proposes a unique way to underhand the age of the contents of a repo, we seek to improve his designs by evenly spacing out various layers of the graph (instead of stacking them), to reduce visual distortions oscillating streams of data.

0.3 Related research papers

- Githru Visual Analytics for Understanding Software Development History Through Git Metadata Analysis
- Git-Truck: Hierarchy-Oriented Visualization of Git Repository Evolution — Git-Truck is a hierarchical file-organized-oriented visualization tool that represents git repositories as different treemaps. Each treemap has its nodes sized by the number of lines of code in the repository, and the color of its nodes can represent a variety of interested factors such as number of commits, top contributors, or even the last change day. This is showing the current structure of a repository, rather than the temporal focus we have on in this project. Mainly, we emphasize the importance of seeing the historical development of features in a repository.
- ChangePrism:Visualizing the Essence of Code Changes

0.4 Planned technical approach

Our visualizations will use the '.git' files for the repos we are studying as a standardized source of data. We plan to implement our visualizations in Javascript, we may pull in dependencies for creating charts as needed.

0.5 Selected dataset(s)

We will focus on three categories of repos in this project.

- Large repositories with many contributes like the Linux kernal
- Large repositories with few contributes like the emacs
- Small repositories with few contributes like mods and web applications created by small teams or individuals. For fun, we will even feed the git repository for this project into our system.

0.6 Expected results and evaluation methodology

In this project, we seek to create a web application to demonstrate and explain our visual analytic systems. It will feature the visualizations we've created, a short text description of the visualizations—explaining works inspiring the visualization, the types of analytical questions the visualization seeks to answer and the strengths and weakness of our approach.

The app will also feature a selector that will allow you to view any git repository with any of the aforementioned visualization techniques.